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Open-Source & Open-Weight Coding Models: Comprehensive Guide (March 2026)

State of the open-source coding model landscape as of March 25, 2026. Covers every notable model family, from frontier-class to laptop-friendly.

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Tier 1: Frontier Open Models

These models compete with GPT-4o / Claude Sonnet 4 / Gemini 2.5 Pro on coding tasks.

1. DeepSeek-V3.1 / V3.2

Field	Details
Creator	DeepSeek AI (China)
Release	V3: Dec 25, 2024; V3.1: Aug 2025; V3.2: Dec 2025
License	MIT
Parameters	671B total / 37B active (MoE)
Context Window	128K tokens
Training Data	14.8T tokens; pre-trained from scratch on diverse code + text
Architecture	Mixture-of-Experts with Multi-head Latent Attention (MLA)

Benchmark Scores: - HumanEval: 85.7% - MBPP: 75.4% - LiveCodeBench: 45.8-49.2% (V3.1) - SWE-bench Verified: ~49.2% (V3), improved in V3.1/V3.2 - Codeforces: Competitive-level performance

Key Strengths: - Exceptional cost-efficiency (trained for ~\$5.5M) - Strong agentic/tool-use capabilities in V3.1+ - Unified thinking/non-thinking mode in V3.1 - Best open model for general-purpose coding + reasoning hybrid tasks - Supports 100+ programming languages

How to Run: Ollama (deepseek-v3), vLLM, llama.cpp (GGUF), HuggingFace TGI. Requires ~350GB VRAM for full precision; quantized versions fit on 4x A100 or 2x H100.

2. Qwen3-Coder (Alibaba)

Field	Details
Creator	Qwen Team, Alibaba Cloud
Release	July 22, 2025
License	Apache 2.0
Parameters	480B total / 35B active (MoE)
Context Window	256K native, 1M with extrapolation
Training Data	7.5T tokens with 70% code ratio

Benchmark Scores: - SWE-bench Verified: ~70% (state-of-the-art among open models at release) - SWE-bench Pro: 44.3% - LiveCodeBench: Strong (specific score varies by evaluation window) - Aider Polyglot: Leading performance

Key Strengths: - Purpose-built for agentic coding (not a general model adapted for code) - 256K native context – handles entire repositories - Comes with Qwen Code CLI tool (open-source terminal coding agent) - Best open model for SWE-bench style real-world software engineering tasks - 92 programming languages supported

How to Run: Ollama (qwen3-coder), vLLM, HuggingFace, LM Studio. FP8 variant available. Requires significant GPU resources for full model; API access via Together AI, OpenRouter.

3. GLM-5 (Zhipu AI)

Field	Details
Creator	Zhipu AI (China)
Release	February 2026
License	MIT
Parameters	~745B (MoE) / 44B active
Context Window	128K+ tokens
Training Data	Trained entirely on Huawei Ascend chips (no NVIDIA GPUs)

Benchmark Scores: - SWE-bench Verified: 77.8% (highest among open-source at time of release) - Humanity’s Last Exam: 50.4% (beats Claude Opus 4.5) - Industry’s lowest hallucination rate claimed

Key Strengths: - Top SWE-bench Verified score among open models - Trained without NVIDIA hardware (Huawei Ascend ecosystem) - Strong at agentic engineering tasks - Excellent code generation + debugging

How to Run: HuggingFace, vLLM. Available on major Chinese cloud platforms.

4. GLM-4.7 (Zhipu AI)

Field	Details
Creator	Zhipu AI (China)
Release	December 22, 2025
License	Open-source (specific license TBD)
Parameters	~355B (MoE) / 32B active
Context Window	200K tokens

Benchmark Scores: - LiveCodeBench: 84.9% (ahead of Claude Sonnet 4.5) - SWE-bench Verified: 73.8% (highest among open-source at time of release)

Key Strengths: - Extremely cost-effective (\$3/month via API, or free locally) - Matches premium closed-source models on real coding benchmarks - Strong across both competitive programming and real-world SWE tasks

How to Run: Ollama, vLLM, HuggingFace TGI.

5. Kimi K2 / K2.5 (Moonshot AI)

Field	Details
Creator	Moonshot AI (China)
Release	K2: Mid-2025; K2.5: Late 2025
License	Modified MIT License
Parameters	1T total / 32B active (MoE)
Context Window	128K tokens
Training Data	15.5T tokens (K2 base); K2.5 adds ~15T mixed visual/text tokens

Benchmark Scores: - SWE-bench Verified: 65.8% (K2) - SWE-bench Multilingual: 47.3% - HumanEval: 99.0% (K2.5 – highest of any model tracked)

Key Strengths: - Top open-source non-reasoning model for coding - Largest open MoE model (1T parameters) - Specifically designed for tool use, reasoning, and autonomous problem-solving - K2.5 excels at front-end development - Zero training instability reported

How to Run: HuggingFace, vLLM. Model weights available for self-hosting.

6. Devstral 2 (Mistral AI)

Field	Details
Creator	Mistral AI (France)
Release	Late 2025
License	Modified MIT
Parameters	123B dense transformer
Context Window	256K tokens

Benchmark Scores: - SWE-bench Verified: 72.2%

Key Strengths: - Dense architecture (not MoE) – simpler deployment - Up to 7x more cost-efficient than Claude Sonnet at real-world tasks - Purpose-built for code agents - Strong at multi-file codebase exploration and editing - Comes with Mistral Vibe CLI (open-source coding assistant)

How to Run: HuggingFace, vLLM, compatible with OpenAI API format.

7. Llama 4 Maverick (Meta)

Field	Details
Creator	Meta AI
Release	April 2025
License	Llama 4 Community License (free up to 700M MAU)
Parameters	~400B total / 17B active (128 experts MoE)
Context Window	1M tokens (instruct), 256K pre-trained
Training Data	Massive multimodal corpus

Benchmark Scores: - LiveCodeBench: 43.4% - Meta claims competitive with GPT-4o - Independent testing (Rootly): 70% accuracy on SRE-focused coding (vs 90% for Qwen2.5-Coder-32B)

Key Strengths: - Natively multimodal (text + image) - 1M context window – largest among open models at release - Only 17B active parameters for inference efficiency - Distilled from Llama 4 Behemoth (2T parameter teacher model)

Caveats: Independent benchmarks have shown underperformance vs Meta’s claims on specialized coding tasks. Not a coding-specialized model.

How to Run: Ollama, vLLM, llama.cpp, HuggingFace. Fits on 2x A100-80GB with quantization.

Tier 2: Strong Open Models

Compete with GPT-3.5 Turbo / Claude Sonnet 3.5 level on coding.

8. Qwen2.5-Coder-32B (Alibaba)

Field	Details
Creator	Qwen Team, Alibaba Cloud
Release	November 2024
License	Apache 2.0
Parameters	32B dense
Context Window	128K tokens
Training Data	5.5T tokens of code data, 92 programming languages

Benchmark Scores: - HumanEval: 92.7% - MBPP: 90.2% - MATH: 57.2% - LiveCodeBench: 31.4% - Aider Polyglot: 73.7% (comparable to GPT-4o) - McEval (multilingual): 65.9 - MdEval: 75.2 (first among open-source)

Key Strengths: - Best pure coding model at the 32B scale - Exceptional multilingual coding (40+ languages) - Strong at code completion, generation, debugging, and translation - Practical to run on consumer hardware with quantization - State-of-the-art open-source code model at release

How to Run: Ollama (qwen2.5-coder:32b), vLLM, llama.cpp, LM Studio, HuggingFace. Fits on single A100-80GB or 64GB Mac with Q4 quantization.

9. DeepSeek-Coder-V2 (DeepSeek AI)

Field	Details
Creator	DeepSeek AI
Release	June 17, 2024
License	Permissive (research + commercial)

Field	Details
Parameters	236B total / 21B active (MoE)
Context Window	128K tokens
Training Data	Code-focused + math enriched dataset

Benchmark Scores: - HumanEval: 90.2% - MultiPL-E: 72.8% average across all languages - Strong math/reasoning capabilities

Key Strengths: - First open model to match closed-source models on code intelligence - Extended context from 16K to 128K via Yarn technique - Strong multilingual code support (338 languages) - Efficient inference with only 21B active parameters

How to Run: Ollama, vLLM, HuggingFace. Smaller active parameter count makes it more efficient than total size suggests.

10. Codestral 25.01 / 25.08 (Mistral AI)

Field	Details
Creator	Mistral AI (France)
Release	Codestral: May 2024; 25.01: Jan 2025; 25.08: Aug 2025
License	Mistral AI Non-Production License (MNPL) for original; varies by version
Parameters	22B
Context Window	256K tokens
Training Data	80+ programming languages

Benchmark Scores: - HumanEval: 86.6% - MBPP: 91.2% - #1 on LMsys Copilot Arena leaderboard (25.01)

Key Strengths: - Optimized for low-latency, high-frequency code completion - Fill-in-the-Middle (FIM) support - Code correction and test generation - 2x faster than original Codestral (25.01 improvement) - Excellent for IDE integrations (tab completion)

How to Run: Ollama (codestral), vLLM, HuggingFace. 22B fits on a single GPU with quantization.

11. Llama 3.3 70B (Meta)

Field	Details
Creator	Meta AI
Release	December 2024
License	Llama 3.3 Community License
Parameters	70B dense
Context Window	128K tokens
Training Data	15T tokens

Benchmark Scores: - HumanEval: 88.4% - Near Llama 3.1 405B performance at fraction of compute

Key Strengths: - General-purpose model with strong coding - 128K context window for large codebases - Well-supported across all major inference frameworks - Matches Llama 3.1 405B quality on many tasks despite 5.8x fewer parameters

How to Run: Ollama (llama3.3:70b), vLLM, llama.cpp, HuggingFace. Requires ~40GB VRAM with Q4 quantization.

12. Llama 3.1 405B (Meta)

Field	Details
Creator	Meta AI
Release	July 2024
License	Llama 3.1 Community License
Parameters	405B dense
Context Window	128K tokens
Training Data	15T+ tokens

Benchmark Scores: - HumanEval: 89.0% - MBPP: ~88.6%

Key Strengths: - Largest dense open model with frontier-class coding - Near GPT-4o / Claude 3.5 Sonnet quality
- 128K context - Excellent for distillation teacher

Caveats: Requires massive infrastructure (8x A100-80GB minimum). Largely superseded by MoE models for practical use.

How to Run: vLLM (multi-GPU), HuggingFace TGI. Not practical for local inference.

13. Devstral Small 2 (Mistral AI)

Field	Details
Creator	Mistral AI (France)
Release	December 2025
License	Apache 2.0
Parameters	24B
Context Window	256K tokens

Benchmark Scores: - SWE-bench Verified: 68.0% (strongest open-weight at this size)

Key Strengths: - Strongest open-weight SWE-bench model at the 24B scale - Outscores many 70B-class competitors
- Runs on a single RTX 4090 or Mac with 32GB RAM - Purpose-built for software engineering agent tasks - Multi-file codebase exploration and editing - Vision capabilities (can analyze images/screenshots)

How to Run: Ollama (`devstral`), llama.cpp (GGUF available from Unsloth), vLLM, LM Studio. Single consumer GPU viable.

Tier 3: Small/Efficient Models

Run on laptops and consumer hardware. Under 15B parameters.

14. Qwen3-Coder-Next (Alibaba)

Field	Details
Creator	Qwen Team, Alibaba Cloud
Release	February 2026
License	Apache 2.0
Parameters	80B total / 3B active (ultra-sparse MoE)
Context Window	256K tokens

Benchmark Scores: - SWE-bench Verified: 70.6% (with SWE-Agent), 71.3% (with OpenHands) - SWE-bench Pro: 44.3% - SWE-bench Multilingual: 62.8% - Terminal-Bench 2.0: 36.2% - SecCodeBench: 61.2%

Key Strengths: - Beats DeepSeek-V3.2 on SWE-bench while using 0.4% of active parameters - 10x higher throughput than full-size models for repository tasks - Ultra-sparse design: only 3B parameters active per token - Purpose-built for coding agents - Runs locally on consumer hardware despite 80B total parameters

How to Run: Ollama (`qwen3-coder-next`), vLLM, LM Studio, llama.cpp. The 3B active parameter count means it runs efficiently on modest hardware.

15. Qwen2.5-Coder (Alibaba) – Smaller Sizes

Field	Details
Creator	Qwen Team, Alibaba Cloud
Release	November 2024
License	Apache 2.0 (0.5B/1.5B/7B/14B/32B); Qwen Research License (3B)
Sizes	0.5B, 1.5B, 3B, 7B, 14B
Context Window	128K tokens
Training Data	5.5T tokens, 92 languages

Benchmark Scores (7B): - HumanEval: 88.4% - Strong across 40+ languages

Benchmark Scores (14B): - Competitive with much larger general-purpose models on code tasks

Key Strengths: - Best coding performance per parameter at every size class - 7B model rivals many 33B models from 2023-2024 - Full range of sizes for every hardware tier - Supports code generation, completion, debugging, explanation

How to Run: Ollama (`qwen2.5-coder:7b`, `qwen2.5-coder:14b`, `qwen2.5-coder:1.5b`), llama.cpp, LM Studio, vLLM.

16. Phi-4 (Microsoft)

Field	Details
Creator	Microsoft Research
Release	December 2024
License	MIT
Parameters	14B
Context Window	16K tokens
Training Data	Synthetic data heavy; curated web + code

Benchmark Scores: - HumanEval: 82.6% - Surpasses Llama 3.3 70B on some benchmarks despite being 5x smaller

Key Strengths: - Exceptional parameter efficiency - Strong math + code reasoning - MIT licensed – fully permissive - 14B fits on consumer GPUs easily

How to Run: Ollama (`phi4`), llama.cpp, HuggingFace, vLLM. Runs on 8GB+ VRAM GPUs.

17. Phi-3 (Microsoft)

Field	Details
Creator	Microsoft Research
Release	April 2024
License	MIT
Sizes	Mini (3.8B), Small (7B), Medium (14B)
Context Window	4K or 128K variants

Benchmark Scores: - HumanEval (Mini): 59.1-62.8% - HumanEval (MoE/3.5): 70.7% - MBPP (MoE/3.5): 80.8%

Key Strengths: - Phi-3 Mini runs on phones (3.8B) - 128K context variant available - Strong for size – rivaling Mixtral 8x7B at a fraction of parameters - MIT licensed

How to Run: Ollama, llama.cpp, ONNX Runtime, vLLM. Phone-deployable for Mini.

18. Yi-Coder (01.AI)

Field	Details
Creator	01.AI (China)
Release	September 2024
License	Apache 2.0
Sizes	1.5B, 9B
Context Window	128K tokens

Benchmark Scores: - HumanEval (9B-Chat): 85.4% - MBPP (9B-Chat): 73.8% - LiveCodeBench (9B-Chat): 23.4% (only sub-10B model to exceed 20%) - CRUXEval-O: >50% (first open-source LLM to achieve this)

Key Strengths: - State-of-the-art coding under 10B parameters - 128K context window – rare at this size - Strong code editing and long-context comprehension - Math reasoning via program-aided settings: 70.3% average accuracy

How to Run: Ollama, llama.cpp, vLLM, HuggingFace. 9B fits on consumer GPUs easily.

19. CodeGemma (Google)

Field	Details
Creator	Google DeepMind
Release	April 2024
License	Gemma license (permissive, commercial allowed)
Sizes	2B (pretrained), 7B (pretrained), 7B (instruction-tuned)
Context Window	8K tokens
Training Data	500B additional tokens of code on top of Gemma base

Benchmark Scores: - HumanEval (2B): 44.5% - HumanEval (7B-IT v1.1): 60.4% - HumanEval Infilling (7B): 76.09% single-line, 58.44% multi-line

Key Strengths: - Excellent code infilling (FIM) capabilities - Lightweight – 2B runs on edge devices - Good for IDE code completion - Multi-language support (Python, C++, Java, Go, JS, Kotlin, Rust)

Caveats: 8K context is limiting. Superseded by newer models on raw benchmarks.

How to Run: Ollama (codegemma), llama.cpp, HuggingFace, NVIDIA NIM.

20. OpenCoder (OpenCoder Team)

Field	Details
Creator	OpenCoder Consortium (multi-institution)
Release	November 2024
License	Apache 2.0
Sizes	1.5B, 8B
Context Window	8K tokens
Training Data	2.5T tokens (1.5B-32B range); full data pipeline open-sourced

Benchmark Scores: - HumanEval (8B-Instruct): ~83.5% - HumanEval (8B-Base): 64.6% - Competitive on BigCodeBench, LiveCodeBench, MultiPL-E

Key Strengths: - Fully reproducible – all training data, code, and recipes open-sourced - 4.5M+ SFT entries released - Best option for research on code LLM training - Supports English and Chinese

How to Run: Ollama (opencoder), llama.cpp, vLLM, HuggingFace.

21. StarCoder2 (BigCode / HuggingFace / ServiceNow / NVIDIA)

Field	Details
Creator	BigCode Project (HuggingFace, ServiceNow, NVIDIA consortium)
Release	February 2024
License	BigCode OpenRAIL-M v1
Sizes	3B, 7B, 15B
Context Window	16K tokens (sliding window attention: 4K)
Training Data	3.3-4.3T tokens from The Stack v2 (600+ languages)

Benchmark Scores: - HumanEval (15B-Instruct): 72.6% - HumanEval (15B-Base): 46.3% - StarCoder2-3B outperforms StarCoderBase-15B - StarCoder2-15B matches/outperforms CodeLlama-34B

Key Strengths: - 600+ programming languages (broadest language coverage) - Permissive OpenRAIL license with responsible use guidelines - Strong on low-resource languages - Part of the most transparent open-source code LLM project - Good math and code reasoning despite moderate size

How to Run: Ollama (starcoder2), llama.cpp, vLLM, HuggingFace. 15B fits on single consumer GPU.

22. Granite Code Models (IBM)

Field	Details
Creator	IBM Research
Release	May 2024 (original); Granite 3.0: Oct 2024; 3.3: Early 2025
License	Apache 2.0
Sizes	3B, 8B, 20B, 34B
Context Window	8K (base), 128K (long-context variants)
Training Data	116 languages; 3B/8B: 4T tokens; 20B: 3T tokens; 34B: 1.4T tokens

Benchmark Scores: - HumanEval (Granite-3.0-8B): 52.44% - Granite-3B is best performing small model (+3% over CodeGemma-2B) - Strong on HumanEvalSynthesize (multilingual: JS, Java, Go, C++, Rust)

Key Strengths: - Enterprise-focused with compliance and safety features - Full Apache 2.0 license – no restrictions - Long-context variants (128K) for repository-level tasks - Code synthesis, fixing, explanation, editing, and translation - Well-integrated with IBM watsonx platform

How to Run: Ollama, vLLM, HuggingFace, IBM watsonx.ai. Also available on AWS SageMaker.

23. WizardCoder (WizardLM Team)

Field	Details
Creator	WizardLM Team (Microsoft Research affiliated)
Release	V1.0: June 2023; V1.1: January 2024
License	Llama 2 Community License (based on Code Llama/StarCoder)
Sizes	15B, 33B, 34B
Context Window	16K tokens

Benchmark Scores: - HumanEval (33B-V1.1): 79.9% - HumanEval+ (33B-V1.1): 73.2% - MBPP (33B-V1.1): 78.9% - Outperforms ChatGPT 3.5, Gemini Pro, DeepSeek-Coder-33B-Instruct

Key Strengths: - Evol-Instruct methodology for code (ICLR 2024 paper) - Strong Python focus - Good for straightforward code generation tasks

Caveats: No updates since January 2024. Superseded by newer models. Limited language breadth.

How to Run: Ollama, llama.cpp, HuggingFace. 33B fits on single A100 or 64GB Mac.

24. Phind-CodeLlama-34B-v2 (Phind)

Field	Details
Creator	Phind
Release	Late 2023
License	Llama 2 Community License
Parameters	34B
Context Window	16K tokens
Training Data	1.5B tokens of high-quality programming instruction-answer pairs

Benchmark Scores: - HumanEval: 73.8%

Key Strengths: - Strong multi-language proficiency (Python, C/C++, TypeScript, Java) - Instruction-tuned for practical coding tasks - Fine-tuned on high-quality problem-solution pairs (not just code completion)

Caveats: No updates since 2023. Superseded by newer models.

How to Run: Ollama, llama.cpp, HuggingFace.

25. Code Llama (Meta)

Field	Details
Creator	Meta AI
Release	7B/13B/34B: Aug 2023; 70B: Jan 2024
License	Llama 2 Community License
Sizes	7B, 13B, 34B, 70B
Context Window	16K trained, up to 100K effective
Training Data	500B tokens (7B-34B); 1T tokens (70B)

Benchmark Scores: - HumanEval (7B): 37.3% - HumanEval (13B): 42.9% - HumanEval (34B): 53.7% - HumanEval (70B): 67.8% - MBPP (70B): 62.2%

Key Strengths: - Foundation model for many fine-tuned variants (WizardCoder, Phind, etc.) - Fill-in-the-Middle support (7B, 13B, 70B) - Python-specialized and instruction-tuned variants available - Well-supported ecosystem

Caveats: Superseded by Llama 3.x and specialized coding models. Lower benchmarks than modern alternatives.

How to Run: Ollama, llama.cpp, vLLM, HuggingFace. Broadly supported.

26. Snowflake Arctic

Field	Details
Creator	Snowflake
Release	April 2024
License	Apache 2.0
Parameters	480B total / 17B active (Dense-MoE Hybrid)
Context Window	4K tokens
Training Data	Enterprise-focused corpus

Benchmark Scores: - HumanEval+: 64.3% - MBPP+: Competitive - Spider (SQL): 79%

Key Strengths: - Best-in-class for SQL generation - Enterprise intelligence focus (SQL, code, instruction following) - Efficient MoE architecture (17B active) - Apache 2.0 with all data recipes open-sourced

Caveats: Small context window (4K). Not competitive with modern coding models on general programming. SQL is the sweet spot.

How to Run: HuggingFace, vLLM. Designed for Snowflake platform integration.

Tier 4: Reasoning Models for Coding

Extended thinking / chain-of-thought models that excel at complex coding problems.

27. DeepSeek-R1 (DeepSeek AI)

Field	Details
Creator	DeepSeek AI
Release	January 2025; R1-0528: May 2025
License	MIT
Parameters	685B total / ~37B active (MoE)
Context Window	128K tokens

Benchmark Scores: - HumanEval: 90.2% - SWE-bench Verified: 49.2% (original), 57.6% (R1-0528) - LiveCodeBench: 63.5% (original), 73.3% (R1-0528) - Codeforces: ~1930 rating (R1-0528)

Distilled Variants (all MIT licensed): | Model | Base | LiveCodeBench | Codeforces | |—|—|—|—| | R1-Distill-Qwen-1.5B | Qwen-2.5-1.5B | 16.9% | — | | R1-Distill-Qwen-7B | Qwen-2.5-7B | Moderate | — | | R1-Distill-Qwen-14B | Qwen-2.5-14B | Good | — | | R1-Distill-Qwen-32B | Qwen-2.5-32B | 57.2% | 1691 | | R1-Distill-Llama-8B | Llama-3.1-8B | Moderate | — | | R1-Distill-Llama-70B | Llama-3.3-70B | 57.5% | 1633 |

Key Strengths: - Chain-of-thought reasoning produces step-by-step solutions - Trained via reinforcement learning (not just SFT) - Matches OpenAI o1 on coding and math - Six distilled variants cover every hardware tier - Thinking process is transparent and inspectable

How to Run: Ollama (deepseek-r1, or distilled: deepseek-r1:7b, etc.), vLLM, llama.cpp, HuggingFace. Distilled 7B/8B run on consumer hardware.

28. QwQ-32B (Alibaba)

Field	Details
Creator	Qwen Team, Alibaba Cloud
Release	March 2025
License	Apache 2.0
Parameters	32B dense
Context Window	128K tokens

Benchmark Scores: - Matches DeepSeek-R1 and OpenAI o1-mini on reasoning - Strong on LiveCodeBench and CodeForces - 32B params vs DeepSeek-R1’s 671B – an order of magnitude smaller

Key Strengths: - Best reasoning-per-parameter ratio - Deep thinking capabilities for complex algorithmic problems - Runs on much more accessible hardware than DeepSeek-R1 - Apache 2.0 license – fully permissive

How to Run: Ollama (qwq), vLLM, llama.cpp, HuggingFace. Fits on single A100 or 64GB Mac.

29. Qwen3 (with Thinking Mode)

Field	Details
Creator	Qwen Team, Alibaba Cloud
Release	April 29, 2025
License	Apache 2.0
Sizes (Dense)	0.6B, 1.7B, 4B, 8B, 14B, 32B
Sizes (MoE)	30B-A3B (30B total/3B active), 235B-A22B (235B total/22B active)
Context Window	128K tokens

Benchmark Scores: - Qwen3-235B: Leads on CodeForces ELO, LiveCodeBench v5, BFCL - Qwen3-30B-A3B: Strong performer, outranked only by QwQ-32B - Thinking mode enables step-by-step reasoning for complex problems

Key Strengths: - Dual mode: Thinking (deep reasoning) and Non-Thinking (quick answers) - 8 sizes covering every hardware tier - MoE variants are extremely efficient - Strong coding, math, and general reasoning - Apache 2.0 for all sizes

How to Run: Ollama (qwen3), vLLM, llama.cpp, HuggingFace, LM Studio. Qwen3-4B runs on phones; 30B-A3B runs on laptops.

30. Phi-4-Reasoning (Microsoft)

Field	Details
Creator	Microsoft Research
Release	April 2025
License	MIT
Parameters	14B
Context Window	32K tokens

Benchmark Scores: - 25+ percentage point improvement on LiveCodeBench over base Phi-4 - Strong on HumanEvalPlus - Competitive with much larger reasoning models

Key Strengths: - Smallest true reasoning model (14B) - Trained via SFT on o3-mini reasoning demonstrations - Reasoning-plus variant available for enhanced performance - MIT licensed – fully open - Runs on consumer GPUs

How to Run: Ollama (phi4-reasoning), llama.cpp, vLLM, HuggingFace.

Benchmark Comparison Tables

Table 1: HumanEval Pass@1 (Python Code Generation)

Model	Size (Active)	HumanEval	MBPP
Kimi K2.5	32B	99.0%	–
Qwen2.5-Coder-32B	32B	92.7%	90.2%
DeepSeek-Coder-V2	21B active	90.2%	–
DeepSeek-R1	37B active	90.2%	–
Llama 3.1 405B	405B	89.0%	~88.6%
Qwen2.5-Coder-7B	7B	88.4%	–
Llama 3.3 70B	70B	88.4%	–
Codestral 25.01	22B	86.6%	91.2%
DeepSeek-V3	37B active	85.7%	75.4%
Yi-Coder-9B-Chat	9B	85.4%	73.8%
OpenCoder-8B-Instruct	8B	~83.5%	–
Phi-4	14B	82.6%	–
WizardCoder-33B-V1.1	33B	79.9%	78.9%
Phind-CodeLlama-34B-v2	34B	73.8%	–
StarCoder2-15B-Instruct	15B	72.6%	–
Phi-3.5-MoE	16B active	70.7%	80.8%
Code Llama 70B	70B	67.8%	62.2%
Snowflake Arctic	17B active	64.3%	–
CodeGemma-7B-IT	7B	60.4%	–
Granite-3.0-8B	8B	52.44%	–
CodeGemma-2B	2B	44.5%	–

Table 2: SWE-bench Verified (Real-World Software Engineering)

Model	Size (Active)	SWE-bench Verified
GLM-5	44B active	77.8%
GLM-4.7	32B active	73.8%
Devstral 2	123B dense	72.2%
Qwen3-Coder-Next	3B active	71.3%
Qwen3-Coder-480B	35B active	~70%

Model	Size (Active)	SWE-bench Verified
Devstral Small 2	24B dense	68.0%
Kimi K2	32B active	65.8%
DeepSeek-R1-0528	37B active	57.6%
DeepSeek-V3	37B active	~49.2%
Devstral (original)	24B	46.8%

Table 3: LiveCodeBench (Competitive Programming)

Model	Size (Active)	LiveCodeBench
GLM-4.7	32B active	84.9%
DeepSeek-R1-0528	37B active	73.3%
DeepSeek-R1	37B active	63.5%
DeepSeek-R1-Distill-Llama-70B	70B	57.5%
DeepSeek-R1-Distill-Qwen-32B	32B	57.2%
DeepSeek-V3.1	37B active	45.8-49.2%
Llama 4 Maverick	17B active	43.4%
Codestral 25.01	22B	37.9%
Qwen2.5-Coder-32B	32B	31.4%
Yi-Coder-9B-Chat	9B	23.4%
DeepSeek-R1-Distill-Qwen-1.5B	1.5B	16.9%

Size vs Performance Analysis

Best Coding Performance Per VRAM Budget

VRAM Budget	Best Model	Key Metric
4-6 GB	Qwen2.5-Coder-1.5B (Q4) or Qwen3-0.6B	Basic completion, 44-50% HumanEval
8 GB	Qwen2.5-Coder-7B (Q4)	88.4% HumanEval – exceptional for the VRAM
12-16 GB	Yi-Coder-9B or Phi-4 (Q4/Q6)	85.4% / 82.6% HumanEval
16-24 GB	Devstral Small 2 (Q4) or Qwen2.5-Coder-14B	68% SWE-bench / strong HumanEval
24-32 GB	Qwen2.5-Coder-32B (Q4) or QwQ-32B (Q4)	92.7% HumanEval or reasoning mode
32-48 GB	Qwen3-Coder-Next (80B, 3B active)	71.3% SWE-bench Verified with only 3B active
48-80 GB	Llama 3.3 70B (Q4) or DeepSeek-R1-Distill-Llama-70B	88.4% HumanEval or 57.5% LiveCodeBench
2x A100 (160GB)	DeepSeek-V3.1 (quantized)	Full frontier performance
4x A100+ (320GB+)	Qwen3-Coder-480B or GLM-5	State-of-the-art SWE-bench

Efficiency Champions (Performance / Active Parameters)

1. **Qwen3-Coder-Next** – 71.3% SWE-bench with only 3B active parameters. The clear winner.
2. **Qwen2.5-Coder-7B** – 88.4% HumanEval at just 7B parameters.
3. **Phi-4** – 82.6% HumanEval at 14B. Best dense model efficiency.
4. **Llama 4 Maverick** – 43.4% LiveCodeBench with 17B active parameters.
5. **DeepSeek-V3.1** – Frontier performance with 37B active out of 671B total.

Recommendations by Use Case

Code Completion in IDE (Tab Autocomplete)

Best: Codestral 25.01 (22B) – purpose-built for FIM, low-latency, #1 on copilot arena. **Budget:** Qwen2.5-Coder-7B or CodeGemma-7B for FIM. **Edge:** Qwen2.5-Coder-1.5B or CodeGemma-2B.

Agentic Coding (Multi-File, Autonomous Engineering)

Best: Qwen3-Coder-480B or GLM-5 – highest SWE-bench scores. **Efficient:** Devstral Small 2 (24B) or Qwen3-Coder-Next (80B/3B active). **API:** DeepSeek-V3.1 (cheapest frontier-class API).

Competitive Programming / Algorithmic Challenges

Best: DeepSeek-R1-0528 or GLM-4.7 – highest LiveCodeBench / Codeforces scores. **Local:** QwQ-32B or DeepSeek-R1-Distill-Qwen-32B.

Code Review & Explanation

Best: DeepSeek-V3.1 (thinking mode for thorough analysis). **Local:** Qwen2.5-Coder-32B or Llama 3.3 70B.

SQL Generation

Best: Snowflake Arctic (79% Spider) or DeepSeek-V3.1. **Small:** Granite Code 8B (enterprise-oriented).

Running on a Laptop (No GPU)

Best: Qwen2.5-Coder-7B (Q4_K_M) – runs well on 16GB RAM Macs. **Smaller:** Phi-4 (Q4), Yi-Coder-9B (Q4), or Qwen3-4B. **Tiny:** Qwen2.5-Coder-1.5B or Qwen3-0.6B for machines with 8GB RAM.

Enterprise / Production Deployment

Best: Granite Code (Apache 2.0, IBM support) or Qwen2.5-Coder (Apache 2.0). **Compliance-sensitive:** Granite Code – designed with enterprise guardrails.

Research & Reproducibility

Best: OpenCoder – full training pipeline, data, and checkpoints open-sourced. **Also:** StarCoder2 – transparent data sourcing via The Stack v2.

Front-End / Web Development

Best: Kimi K2.5 – specifically noted for front-end development excellence. **Also:** Qwen3-Coder or Devstral Small 2.

Multi-Language Support (Non-Python)

Best: StarCoder2-15B (600+ languages) or Qwen2.5-Coder-32B (92 languages). **Also:** Codestral (80+ languages), DeepSeek-Coder-V2 (338 languages).

How to Run These Models

Ollama (Easiest Local Setup)

```
# Install Ollama
curl -fsSL https://ollama.com/install.sh | sh

# Run models
ollama run qwen2.5-coder:32b
ollama run deepseek-r1:32b
```

```
ollama run codestral
ollama run qwen3-coder
ollama run qwen3-coder-next
ollama run devstral
ollama run phi4
ollama run yi-coder:9b
ollama run starcoder2:15b
ollama run opencoder:8b
ollama run deepseek-v3 # Requires significant RAM
```

llama.cpp (Maximum Performance, Custom Quantization)

```
# Build llama.cpp
git clone https://github.com/ggerganov/llama.cpp
cd llama.cpp && make -j

# Download GGUF models from HuggingFace (e.g., from TheBloke, Unsloth)
# Run with custom settings
./llama-server -m model.gguf -c 8192 -ngl 99 --port 8080
```

vLLM (Production Serving)

```
pip install vllm

# Serve a model with OpenAI-compatible API
python -m vllm.entrypoints.openai.api_server \
  --model Qwen/Qwen2.5-Coder-32B-Instruct \
  --tensor-parallel-size 1 \
  --max-model-len 32768
```

HuggingFace Transformers

```
from transformers import AutoTokenizer, AutoModelForCausalLM

model_name = "Qwen/Qwen2.5-Coder-7B-Instruct"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForCausalLM.from_pretrained(model_name, device_map="auto")
```

LM Studio (GUI for Mac/Windows/Linux)

Download from lmstudio.ai. Search for models directly in the app. Supports GGUF quantized models with one-click download and run.

Tool Compatibility Matrix

Aider (AI Pair Programming)

Model	Support Level	Notes
DeepSeek-V3.1	Excellent	Via API; 73.7% on Aider benchmark
Qwen2.5-Coder-32B	Excellent	Via Ollama or API; top Aider Polyglot score
Qwen3-Coder	Excellent	Via API or local
Llama 3.3 70B	Good	Via Ollama
Codestral	Good	Via Mistral API
Any Ollama model	Good	Aider supports <code>--model ollama/model-name</code>

Continue (VS Code / JetBrains Extension)

Model	Support Level	Notes
All Ollama models	Native	Configure in <code>config.json</code>
All OpenAI-compatible APIs	Native	vLLM, llama.cpp server, LM Studio
Codestral	Excellent	Purpose-built for tab completion in Continue
Qwen2.5-Coder	Excellent	Recommended for autocomplete

Cline (VS Code Extension – Autonomous Coding Agent)

Model	Support Level	Notes
All Ollama models	Supported	Adjust context window to match VRAM
Devstral Small 2	Excellent	Specifically designed for Cline-style agents
Qwen3-Coder-Next	Excellent	Optimized for agentic coding tools
DeepSeek-V3.1	Excellent	Via OpenAI-compatible API
Any OpenAI-compatible API	Supported	LM Studio, vLLM, llama.cpp

Claude Code (via API Proxy)

Model	Support Level	Notes
DeepSeek-V3.1	Usable	Via OpenAI-compatible proxy (e.g., LiteLLM)
Qwen3-Coder	Usable	Via proxy with function calling support
Any model with tool-use	Experimental	Requires proxy that translates tool-calling format

Cursor / Windsurf / Other IDEs

Most IDEs support OpenAI-compatible APIs, so any model served via vLLM, llama.cpp, or LM Studio can be used. Performance varies based on the model’s function-calling and code-completion capabilities.

Notable Trends (March 2026)

1. **MoE dominance:** Nearly every frontier model uses Mixture-of-Experts. Active parameter counts of 3B-37B provide frontier performance at a fraction of dense model cost.
2. **SWE-bench is the new HumanEval:** HumanEval saturation (99%+ scores from Kimi K2.5) has shifted focus to SWE-bench Verified as the primary coding benchmark. Real-world repository-level tasks matter more than isolated function generation.
3. **Chinese labs lead open-source:** DeepSeek, Qwen (Alibaba), Zhipu AI (GLM), Moonshot AI (Kimi), and 01.AI (Yi) collectively produce the majority of top open coding models. Mistral (France) is the notable Western exception.
4. **Agentic coding models:** The latest generation (Qwen3-Coder, Devstral, GLM-4.7) is specifically trained for autonomous software engineering – multi-file editing, tool use, terminal interaction – not just code completion.
5. **Ultra-sparse efficiency:** Qwen3-Coder-Next (80B/3B active) demonstrates that extreme sparsity can achieve near-frontier SWE-bench scores while being runnable on consumer hardware.

6. **Reasoning models for code:** DeepSeek-R1, QwQ, Phi-4-Reasoning, and Qwen3’s thinking mode bring chain-of-thought reasoning to coding, excelling on competitive programming and complex debugging.
7. **Context windows expanding:** 128K is now standard; 256K available on Codestral/Devstral/Qwen3-Coder; 1M available on Llama 4 Maverick and Qwen3-Coder (extrapolated).
8. **Apache 2.0 becoming standard:** Most new releases use Apache 2.0 or MIT, making commercial use straightforward. Restrictive licenses (Llama Community License, MNPL) are becoming the exception.

Quick Reference: Model Release Timeline

Date	Model	Milestone
Aug 2023	Code Llama	Meta’s first dedicated code model
Feb 2024	StarCoder2	600+ language coverage
Apr 2024	CodeGemma	Google’s code-specialized Gemma
Apr 2024	Snowflake Arctic	Enterprise SQL focus
May 2024	Granite Code	IBM’s enterprise code models
Jun 2024	DeepSeek-Coder-V2	First open model to match closed-source
Sep 2024	Yi-Coder	SotA under 10B params
Nov 2024	Qwen2.5-Coder	92.7% HumanEval at 32B
Nov 2024	OpenCoder	Fully reproducible training
Dec 2024	DeepSeek-V3	MoE efficiency breakthrough
Dec 2024	Llama 3.3	70B matching 405B
Dec 2024	Phi-4	82.6% HumanEval at 14B
Jan 2025	DeepSeek-R1	Open reasoning model revolution
Jan 2025	Codestral 25.01	#1 copilot arena
Mar 2025	QwQ-32B	Reasoning at 32B
Apr 2025	Llama 4	MoE + 1M context
Apr 2025	Phi-4-Reasoning	Reasoning at 14B
Apr 2025	Qwen3	8 sizes, thinking mode
May 2025	DeepSeek-R1-0528	57.6% SWE-bench
Jul 2025	Qwen3-Coder-480B	70% SWE-bench, agentic
Aug 2025	DeepSeek-V3.1	Unified thinking/non-thinking
Dec 2025	GLM-4.7	84.9% LiveCodeBench, 73.8% SWE-bench
Dec 2025	Devstral 2 / Small 2	72.2% / 68.0% SWE-bench
Dec 2025	DeepSeek-V3.2	Iterative improvement
Dec 2025	Kimi K2.5	99.0% HumanEval
Feb 2026	GLM-5	77.8% SWE-bench, trained on Ascend
Feb 2026	Qwen3-Coder-Next	71.3% SWE-bench with 3B active
TBD 2026	DeepSeek-V4	1T params, 1M context (not yet released)

Summary

The open-source coding model landscape in March 2026 is remarkably competitive with closed-source alternatives. For practical use:

- **If you need the absolute best open coding model:** GLM-5 (77.8% SWE-bench) or Qwen3-Coder-480B (~70% SWE-bench) for agentic tasks; Kimi K2.5 (99% HumanEval) for raw code generation.
- **If you want frontier quality at reasonable compute:** DeepSeek-V3.1 (37B active) offers the best performance-per-dollar via API.

- **If you need to run locally on a single GPU:** Qwen2.5-Coder-32B (92.7% HumanEval) or Devstral Small 2 (68% SWE-bench at 24B).
- **If you want reasoning for hard problems:** DeepSeek-R1 or QwQ-32B.
- **If you have only a laptop CPU:** Qwen2.5-Coder-7B quantized or Qwen3-Coder-Next (3B active).
- **If reproducibility matters:** OpenCoder (full pipeline open-sourced).
- **If you need SQL specifically:** Snowflake Arctic.
- **If enterprise compliance matters:** IBM Granite Code (Apache 2.0, enterprise guardrails).

The gap between open and closed models has never been smaller. For many practical coding tasks, open models are now the better choice when considering cost, privacy, and customizability.

Document compiled: March 25, 2026 Coverage: All notable open-source and open-weight coding models through March 2026